Segment–Task Specialization Does Not Emerge in Active-Dendrite Networks: A Dead-Neuron Artifact, a Repair That Does Not Transfer, and What the Own-Segment Bias Actually Does

Ranxi Zhou[a]

[a]*Hunan University of Chinese Medicine, 300 Xueshi Avenue, Hanpu, Yuelu District, Changsha 410208, Hunan, China*

Abstract

Active-dendrite architectures route each input through a competition among dendritic segments, one of which is trained per task; usage-localized elastic weight consolidation (DeLP) protects what has been learned by scaling per-neuron Fisher anchors with segment usage. Both mechanisms rest on an implicit assumption that has, to our knowledge, never been measured: that the segment–task mapping constitutes a *specialization* that protection can leverage at sub-neuron granularity. We audit this assumption on Permuted-MNIST and Permuted-Fashion and report six findings, four of them negative, that together invert the interpretation of a repair we had originally reported as causal. **First**, a granularity ladder of anchors (plain, neuron-resolved, segment-resolved with attribution-tracked reference states) shows that the entire benefit of usage localization is realized at neuron level: segment-resolved anchors are statistically indistinguishable from neuron-resolved anchors across every combination of two datasets, two task counts (T=10, 20),

*Corresponding author.
Email address: zhouranxi@foxmail.com (Ranxi Zhou)

and two elastic weights ($|\Delta\ell| \leq 0.24$, all n=10 paired tests n.s.), and remain so under FIFO anchor budgets ($|\Delta\ell| \leq 0.44$ across eight budget cells). **Second**, the audit explains why: task-dominant gating is *absent* — among active neurons the probability that a neuron's dominant segment is its own task's segment starts each phase at chance and is pushed *below* chance by that phase's own training (T=20: $0.05 \rightarrow 0.01$), and Fisher mass is diffuse across segments (top-1 share 11–18%). The audit also uncovers a measurement artifact relevant to the active-dendrite literature: under production sampling budgets 56–63% of neurons are never activated, and naive argmax instrumentation over all neurons fabricates a strong "segment-0 dominance" ($\approx 60\%$) that vanishes under active-only filtering. **Third**, a constant bias b on the own-segment dendritic input appears to repair this: it raises the learning term in every cell of a {dataset} $\times$ {depth} $\times$ {granularity} grid ($\Delta\ell > 0$ in 22/22 dose cells, significant in 16/22 at n=10 paired; $\Delta\ell$ up to +4.6, $t(9)$=8.1). **Fourth**, however, the repair does not transfer. On a convolutional active-dendrite backbone the dose–response collapses to noise (0/12 cells survive correction) and final accuracy reverses (down to -2.29, $t = -3.1$); a same-architecture control separates the candidate causes by showing that mortality and repair gain *decouple* (GroupNorm raises mortality to 76.5%/85.5% while collapsing the gain to 17%/26% of its bare-MLP value), and a 2×2 width $\times$ normalization scan (320 runs) confirms the attribution — normalization collapses the gain at both widths, while widening the trunk 7.8$\times$ *increases* it. **Fifth**, a mechanism probe re-qualifies what the repair does: the winning segment's gate value is a constant of the training run (0.565), identical under every alternative context, so the context path-

way is present but unused as a modulator; at b=10 the own segment always wins with $\sigma(10 + u) \approx 1$, making the network context-degenerate, and a four-arm control (base / constant-gate / repaired / gate-deleted) shows that gate *deletion* reproduces the gain on MNIST and outperforms the repair on Fashion-MNIST. The honest summary is that we measured the size, the cost, and the boundary of the repair precisely, and located its mechanism only far enough to know that it is not the one we first assumed: the bias does not restore routing, it removes a constant damping factor that a never-used gate had left in place. Sixth, the mechanism result has an uncomfortable implication for our own second finding, and we state it rather than leave it for a reviewer: if the gate's value does not depend on which segment wins, then "dominant segment" is not a description of a routing decision but of an argmax over near-degenerate rows. We check what survives — the granularity ladder compares anchor constructions and never uses a routing statistic, the dead-neuron artifact is a property of argmax on a zero block, and the mechanism and ablation measure outputs and interventions — and adopt the weaker formulation that check licenses: we cannot claim that specialization is actively violated as a property of learned computation, only that no evidence of it is detectable by an instrument that is itself unreliable in this family. Segment–task specialization in this architecture family is therefore not a weak tendency but an assumption with no measurable support — and the absence it leaves is compensated for rather than cured.

Keywords: continual learning, elastic weight consolidation, active dendrites, dendritic segments, catastrophic forgetting, interpretability audit, reproducibility, negative results

1. Introduction

Biological neurons integrate thousands of synaptic inputs across morphologically distinct dendritic compartments, and single-neuron studies have long argued that these compartments act as quasi-independent computational subunits [1, 2, 3]. An influential line of work has translated this principle into artificial networks: context signals select one of several dendritic *segments* per neuron, a winner-take-all competition among segments determines the nonlinear contribution to the soma, and *only the selected segment receives gradient* on a given task [4]. The architectural promise is a division of labor: if task k's computation flows predominantly through segment k, then tasks naturally occupy disjoint sub-network state, and interference between tasks has a structural bottleneck.

Continual-learning methods built on dendritic architectures inherit this promise, often silently. DeLP (usage-localized elastic weight consolidation) protects per-neuron parameter blocks with Fisher anchors scaled by how heavily each neuron was used in past tasks [5, 6]. Its protection unit is the neuron; the segment structure below the neuron is used only to define usage. A natural proposal — one we initially expected to succeed — is to push protection *down* to the segment level: if tasks really live in different segments, anchors that track each segment's own reference state should protect more precisely and drag less on future learning, echoing the granularity intuition that finer protection localizes interference better.

This paper reports what actually happens when that assumption is tested rather than assumed. The narrative is a sequence of three failures that invert the interpretation of the last one, and we report it in that order because the

ordering is the finding: an apparent success in the fourth contribution below is not a success, and we say so.

Our contributions are fivefold, and each arose from the failure of the previous one:

1. **A granularity audit with a null result at segment level (Sec. 3).** We implement segment-resolved anchors with *attribution-tracked reference states* — each anchor tracks where its own segment's parameters drift — and compare a three-rung ladder (plain, neuron-resolved, segment-resolved) across two datasets, two depths, and two elastic weights with n=10 paired seeds. The entire benefit of usage localization is realized in the plain→neuron step ($\Delta\ell$ between -2.8 and -5.4, $|t(9)|$ up to 22); the neuron→segment step adds nothing ($|\Delta\ell| \leq 0.24$, all n.s.), under unbounded stacks and under FIFO anchor budgets alike. The null is mechanistically pinned, not merely observed: attribution is *self-limiting* (the anchor's own penalty pins the weights, so the tracked reference never moves) and segment-level parameter ownership lacks a separable carrier.
2. **A measurement artifact with literature-wide relevance (Sec. 4).** Auditing *why* the null occurs, we find that under production sampling budgets 56–63% of neurons are never activated, and that naive dominant-segment instrumentation (argmax over segment activations of *all* neurons) attributes every dead neuron to segment 0 — fabricating an apparent "segment-0 dominance" of $\approx$ 60% that vanishes under active-only filtering. Among active neurons the true picture is *anti-specialization*: each phase's own training pushes the own-segment

win rate from chance to below chance (T=20: 0.05 → 0.01), and anchor Fisher mass is diffuse across segments (top-1 share 11–18%). Any study that measures segment routing with naive argmax — including evaluations of segment specialization claims — is at risk of the same artifact.

3. **An apparent causal repair (Sec. 5).** If missing specialization is load-bearing rather than cosmetic, restoring it should help. It does: adding a constant bias b to the own-segment dendritic input during training, evaluation, and anchor construction raises the learning term in every cell of a {dataset}×{depth}×{granularity} grid — $\Delta\ell$ positive in 22/22 dose cells, significant in 16/22 (n=10 paired) — with final accuracy significant simultaneously in 16/22 and in every grid cell at some dose ($\Delta\ell$ up to +4.6 at T=20, $t(9)$=8.1); forgetting is *reduced* at T=10 (ΔBWT up to +1.7, $t(9)$ up to 10.7) and mildly traded back at T=20 (significant in 3/10 cells) (Sec. 5). The response is saturated and nontrivially mediated: the own-segment win rate saturates at 13–19% regardless of dose. Signed-drift measurement shows this is not adversarial compensation — the network never drifts against the bias — but a scale effect: the raw own gate spans $|u| \leq 0.32$, at least three times below any tested dose, so the bias forces the same training-time routing at every dose, so only a minority of neuron-tasks flip to the own-segment pathway — and that partial restoration already yields most of the gain.

4. **The repair does not transfer, and the reason is not gate death (Sec. 6).** The third contribution is the one we most wanted to be

wrong about. On a convolutional active-dendrite backbone (160 runs) the dose–response collapses to noise (4/12 cells significant uncorrected, 0/12 after Holm correction) and final accuracy *reverses* (Fashion T=10: Δavg $= -2.29$, t=-3.1). The natural explanation — the repair revived dying gates, and conv gates do not die — is testable, and we test it: inserting GroupNorm into the MLP raises mortality to 76.5%/85.5% while collapsing the repair gain to 17%/26% of its bare-MLP value. Mortality and gain *decouple* in both directions (highest mortality with least gain; zero mortality with no gain), so gate death is neither necessary nor sufficient. A 2×2 width $\times$ normalization scan (320 runs) removes the one confound we had to disclose: normalization collapses the gain at both widths, while widening the trunk 7.8$\times$ *increases* it.

5. **What the repair actually repairs (Sec. 11).** A mechanism probe re-qualifies the third contribution. The winning segment's gate value is a constant of the training run (0.565), identical to four decimals under every alternative context, and the dendritic columns stay mutually orthogonal at their random initialization (mean pairwise cosine 0.003): the context pathway is present, protected per task, and *unused as a modulator*. At b=10 the own segment always wins with $\sigma(10+u) \approx 1$, which makes the network context-degenerate (swapping the context one-hot changes 0.007–0.021% of predictions, against 14.8–30.1% for the base network). A four-arm control (base / constant-gate / repaired / gate-deleted, n=10 paired) settles what the bias buys: pinning the gate at the base system's measured constant reproduces the base system (63.62 vs. 63.86 on MNIST, n.s.), while *deleting* the gate

reproduces most of the repair's gain on MNIST (67.50 vs. 68.67) and *outperforms* it on Fashion-MNIST (44.27 vs. 42.89). The bias does not restore routing; it removes a constant damping factor that a never-used gate had left in place.

6. **A self-interrogation we should have run first (Sec. 13).** The mechanism result has an uncomfortable implication for our own second contribution, and we state it rather than leave it for a reviewer: if the gate's value does not depend on which segment wins, then "dominant segment" is not a description of a routing decision but of an argmax over near-degenerate rows. We check what survives. The granularity ladder (Sec. 3) compares anchor constructions and never uses a routing statistic; the dead-neuron artifact is a property of argmax on a zero block; the mechanism and ablation measure outputs and interventions. All three stand. What does *not* survive is the strong form of our second contribution: we cannot claim that specialization is *actively violated* as a property of learned computation, only that no evidence of it is detectable by an instrument that is itself unreliable here. We adopt the weaker formulation throughout.

Scope relative to a companion study.. A separate paper in this line asks a different question about the same system: how large the protection set should be. It treats the anchor stack as the object of study, bounds the accumulated Fisher mass, and derives a log-linear law relating the learning term to the protected-mass horizon ($\ell \approx 84 - 11 \ln h$; per-seed $R^2 \geq 0.98$; replicated at a second task count and on Split-CIFAR-100 with a ResNet-18 backbone). That question and this one are compatible but independent.

Bounding the stack answers how much protection to keep, not whether finer protection would be worth having; and this paper's answer — that protection granularity below the neuron buys nothing, because the sub-neuron structure it would exploit is not there — implies that a bounded stack of neuron-resolved anchors is the right object to bound. Readers who want the account of *how much* to protect should consult that study; this paper is about *what* there is to protect.

Together the six results reframe segment specialization in this architecture family: not a weak tendency to be exploited, but an assumption with no measurable support under a corrected instrument — and whose absence a one-parameter intervention compensates for rather than cures. The last of these results is the one we would most like to have run first, and the reason we did not is worth stating: the apparent repair was strong, consistent, and saturating, and that is exactly what made it seem unnecessary to ask what the gate was doing. All experiments are on Permuted-MNIST and Permuted-Fashion at $T \in \{10, 20\}$ with paired-seed statistics; code, per-run records, analysis scripts, and the pre-registrations are public (Data Availability).

2. Background and setup

2.1. Active-dendrite network

We use the feed-forward active-dendrite network of Iyer et al. [4] in the protected variant established in our prior work [5]. A one-hidden-layer network of N_h neurons receives a one-hot task context $c_t \in \{0, 1\}^T$; each neuron j carries T dendritic segments, and segment k of neuron j computes $u_{j,k} = \sigma(w_{(j,k)}^\top x + b_{j,k} + v_k^\top c_t)$, where $v \in \mathbb{R}^{T \times (n_{seg} N_h)}$ are the context-

to-dendrite weights — *column* k of v gates segment k for every neuron. The soma nonlinearity applies a k-winners-take-all (kWTA) selection over $h_j = f_j(x)\,\sigma(\max_k u_{j,k})$: the k largest somal pre-activities pass; the rest are set to zero. Crucially, task t's gradient updates are masked to the parameters of segment t ("protect_grads"): the architecture's design intent is that task t trains segment t.

2.2. DeLP: usage-localized EWC and the identity decomposition

DeLP [5] protects the feed-forward and output blocks with per-task anchors, each an elastic penalty $\frac{\lambda}{2}\sum_p F_p\,(W_p - \theta_p)^2$ whose Fisher term F is *usage-localized*: per-neuron usage $o_j \in [0,1]$, the frequency with which neuron j is active under task t's data, scales the anchor rows of the hidden weights and columns of the output weights, so that rarely-used neurons are protected lightly and remain available to future tasks. Writing $\bar{a}_T$ for final average accuracy, the decomposition

$$\bar{a}_T \;=\; \ell \;+\; \tfrac{T-1}{T}\,\mathrm{BWT}, \tag{1}$$

separates what each task achieves when it is learned (ℓ, the learning term: the mean over tasks of the accuracy on a task immediately after that task has been trained) from what earlier tasks retain ($\mathrm{BWT} \le 0$); we report all three, with paired t-statistics over n=10 seeds sharing permutation streams. Where a table quotes an accuracy on the *final* task alone, it says so explicitly (Table 2).

2.3. Protocol

Benchmarks are Permuted-MNIST and Permuted-Fashion [7, 8] with $T \in \{10, 20\}$ tasks, one hidden layer of 256 neurons, 2 epochs per task, Adam

(lr 10^{-3}), Fisher/usage samples of 300/500 examples, and $\lambda \in \{1000, 5000\}$ calibrated in prior work. Seeds are paired across arms (identical permutation streams and init); all comparisons are paired t-tests with n=10, and we mark $|t| > 2$ as significant. All hypotheses in Secs. 3 and 5 were preregistered in writing before the corresponding data were collected.

3. Does segment-resolved protection help? A granularity ladder

3.1. Segment-resolved anchors with attribution-tracked references

A segment-resolved anchor must differ from a neuron-resolved one in more than bookkeeping: since $\sum_k o_{t,j,k} = o_{t,j}$ (segment usages partition neuron usage), splitting a neuron's single Fisher scalar across segments while sharing one reference state θ leaves the penalty algebraically identical to neuron level. We therefore give each segment attribution of parameter drift: anchor a (task t, neuron j) maintains per-segment reference states $\theta^{(a,k)}$, and after each subsequent task t' the realized parameter displacement is partitioned across segments by the usage-driven attribution weights of t''s own anchors; the penalty charges anchor a only for the drift not explained by interlopers in its own segment. The design intends that an anchor no longer litigates, on shared weights, drift caused by tasks that never touched its segment. Segment-level Fisher mass $m_{t,j,k}$ and dominant-segment histograms are instrumented per anchor.

3.2. Result: the ladder is flat above neuron level

Table 1 shows the ladder. The plain→neuron step is large and universal: at λ=1000, T=10 MNIST the learning term gains +4.66 ($t(9)$=22.3);

across all six cells of {dataset} × {depth} × {λ}, $\Delta\ell$ ranges from -2.78 to -5.44 in favor of neuron-level localization with $|t(9)| \in [10.4, 22.3]$. The neuron→segment step is empty on the learning axis: the largest deviation across all twelve pairings is $|\Delta\ell| = 0.24$, never significant (largest $|t| = 1.3$); BWT differences are within 0.43 points. Two of those BWT differences are nominally significant (effect sizes below 0.5 points, $|t| \leq 4.0$) and we report them rather than suppress them: they are marginal forgetting effects, not evidence for a granularity effect, since they sit on the stability side rather than the learning side that the pre-registered hypothesis was about.

Single-anchor references (Table 2) frame the arms; their λ values are the calibration carried over from our companion pipeline, not re-tuned per benchmark. Plain EWC buys ℓ at 3–5× the forgetting; SI (λ=100) is over-stable (its ℓ_{final}=13.2 at T=10 shows it barely learns the later tasks at all, even though its mean learning term ℓ=40.0 is not unremarkable); MAS (λ=1) maximizes the learning term ($\ell \geq 85.7$, the highest of all arms) yet posts catastrophic BWT (-32.6 at T=20) — the regime-dependence of warrant documented in our companion work extends from Split-CIFAR-100 to the MLP setting. One comparison the table makes possible is worth stating explicitly: at T=10, the repaired DeLP arm (b=10; avg 68.7, BWT -9.9) dominates MAS (avg 66.6, BWT -21.2) on both summary terms — MAS buys its higher learning term with exactly the forgetting that the repair removes.

3.3. The flatness survives anchor budgets

Because segment-resolved anchors might pay off only when the anchor *stack* is bounded — fewer anchors plausibly sharpen the value of each — we

crossed granularity with FIFO budgets $M \in \{2, 5\}$ at λ=1000 (preregistered: alive iff $\geq 4/6$ cells significant positive and none significantly worse on BWT). The result is the same null, tighter: across all eight $\{\text{dataset}\} \times \{M\} \times \{T\}$ cells the branch$-$neuron $\Delta\ell$ lies within ± 0.44 with no cell significant positive (one cell, Fashion M=2, T=10, is significantly *negative* at -0.33, $t(9) = -2.3$; Table 3). Budgets shift both granularities identically, consistent with the shared mechanism analysis below.

4. Why the ladder is flat: auditing segment–task specialization

The null in Sec. 3 would be easy to misread as "the segment machinery is redundant." The audit says something stronger: the specialization that segment-resolved protection would need does not exist, and part of the evidence that appeared to show it was an artifact.

4.1. A measurement artifact: dead neurons fabricate dominance

Our first-pass instrumentation recorded, for every neuron, the argmax over its segment activations under each task's data. It showed segment 0 "dominating" 58–63% of neurons — a striking positional bias that initially looked like the failure of task-specific routing. The correction: under the production sampling budget (500 examples), 56–63% of neurons are *never activated*; for a dead neuron all segment activations are exactly zero, argmax breaks ties toward index 0, and the neuron is silently booked to segment 0 (Fig. 1). Filtering to active neurons collapses the apparent bias to chance ($\approx 0.10 = 1/T$ at T=10; untrained networks show no positional preference either, 0.096–0.111 over three seeds). The death itself is real and grows with the sequence: the inactive fraction jumps from 0.02 to 0.35 within

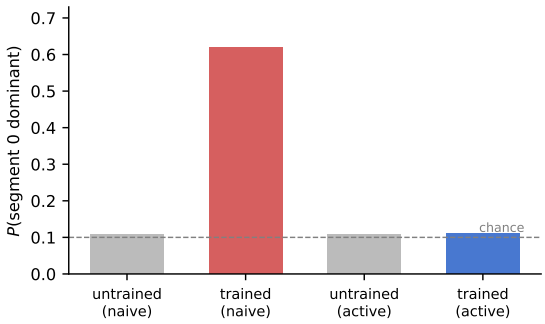


Figure 1: The dead-neuron artifact. Naive argmax over *all* neurons reports segment-0 dominance of $\approx 60\%$ after training; the same statistic over active neurons only, and both statistics on untrained networks, sit at chance ($1/T$).

the first phase of training and stabilizes near 0.6 (Fig. 2) — under kWTA, neurons pushed out of the winner set early are never re-selected by any later task. We flag this as a methodological warning for the active-dendrite literature: dominance statistics computed over all neurons conflate routing with mortality, and at production sampling budgets the conflation is an order of magnitude larger than the signal.

4.2. Anti-specialization among active neurons

With the artifact controlled, the true routing picture is the opposite of specialization. Define p_{own} as the probability that a neuron's dominant segment is the currently measured task's own segment, over active neurons. At the *start* of each phase, p_{own} sits at chance ($1/T$): nothing in the context-gated routing maps tasks to segments. Training within the phase then pushes it *below* chance — by the end of phase p, p_{own} drops to 0.058–0.085 at T=10 (chance 0.10) and, late in a T=20 sequence, to 0.005–0.02 against a chance of 0.05, i.e. five to ten times *below* random assignment (Fig. 3). The own-

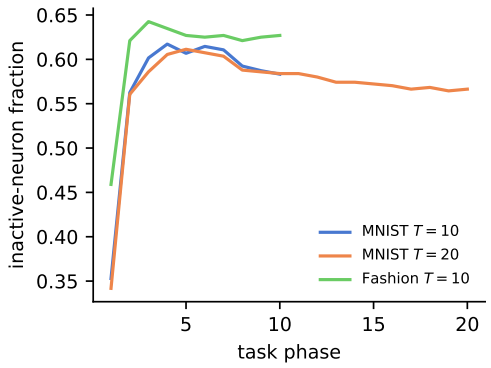


Figure 2: Inactive-neuron fraction across the task sequence (end-of-phase means). Neurons lost in early phases are never recovered; the active substrate available to routing is roughly halved by phase 2 and stabilizes near 40%.

segment gating block does train — it moves by 6–16% of its norm per phase — yet this movement fails to convert into dominance: win rates still end each phase below chance. Segment–task alignment is not merely absent; its absence persists through active training of the very parameters that would encode it.

4.3. Fisher mass is diffuse across segments

Segment-resolved anchors also lack a substrate to hold onto: for trained anchors, the fraction of a neuron's Fisher mass attributable to its top segment is only 11–18% (mean over anchor–neuron pairs). Even a perfectly attributed segment-level penalty would spread nearly all of its protection across segments the anchor task never drove. This measurement closes the mechanistic account of the Sec. 3 null from two sides: attribution is self-limiting (an anchor's own penalty pins the shared weights it measures, so its tracked references barely move), and the per-segment mass that attribution

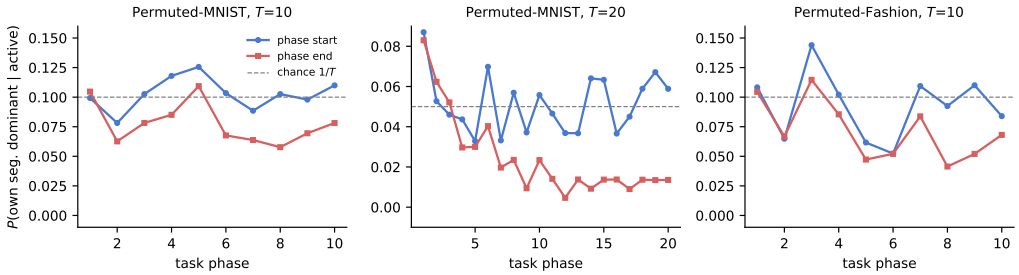


Figure 3: Anti-specialization over training. Within each phase, the own-segment win rate over active neurons starts at chance (dashed) and ends below it; the deficit deepens with sequence position (rightmost points of each curve), most severely at T=20.

would partition is diffuse to begin with.

5. Causal probe: restoring own-segment advantage

How to read this section.. The intervention studied here produces a large, consistent, saturating effect on the learning term, and in the version of this work that first reported it, that effect was taken to be a causal repair of the missing specialization. Two later sections of this manuscript do not sustain that reading: on a convolutional active-dendrite backbone the dose–response collapses to noise (Sec. 6), and a mechanism probe shows that the repaired gate is *context-degenerate* — the bias works by saturating a gate that was never context-dependent, not by restoring routing (Secs. 11–12). We report the dose–response in full anyway, for a reason that only becomes clear later: its *shape* is what rules out the dying-gate explanation. A repair that revived dying gates would be expected to weaken when mortality changes; this one does not, which points elsewhere. A reader who wants only the outcome may skip to Sec. 6; a reader assessing the mechanism should read the dose–

response first.

5.1. A one-parameter intervention

If the missing specialization is load-bearing, causally restoring it should improve the trade-off in Eq. (1). We add a constant bias b to the own-segment dendritic input: $u_{j,t} \mathrel{+}= b$ whenever the context selects task t. The bias is active in training, evaluation, and anchor construction (the Fisher measurement that drives usage-localized protection); the dominant-segment win-rate instrument deliberately reads the raw, bias-free gates, so that every routing statistic in this paper describes what the network itself encodes rather than what the bias forces (Sec. 5.3). The intervention is deliberately dumb — no learned parameters, no schedule, no per-neuron structure — so that any effect is attributable to the routing prior alone. Doses $b \in \{0, 1, 3, 10\}$; all comparisons paired against b=0 on identical seeds.

5.2. Result: the learning term and accuracy improve everywhere

Statistics.. All tests in this section are paired t-tests over $n = 10$ seeds sharing permutation streams and initialization. Because the dose grid spans three metrics across 22 cells, we report Holm-corrected p-values within each metric family ($\Delta\ell$, Δavg, ΔBWT) in addition to the raw values; effect sizes are paired standardized means ($d_z = \bar{d}/s_d$) and intervals are 95% CIs for the paired mean difference. The correction removes two cells from the $\Delta\ell$ family and four from the Δavg family (both on Fashion-MNIST at $T = 10$, where the effect is smallest); it does not touch the qualitative result, which rests on the MNIST cells and on the $T = 20$ monotone dose–response.

Tables 4 and 5 give the dose–response; Fig. 4 plots it. The full grid is 2 datasets $\times$ 2 depths $\times$ 2 granularities $\times$ 3 doses = 24 cells; 22 of them are reported here (the two T=20 branch cells at b=1 were not collected), and across the 22 cells we do report, $\Delta\ell$ is positive in every cell and significant in 16/22; ℓ and final accuracy are simultaneously significant in 16/22, and every grid cell has at least one dose significant on both terms. The effect grows with depth: at T=20 MNIST, b=10 lifts ℓ by +4.56 ($t(9)$=8.1) and final accuracy by +3.81 ($t(9)$=10.6) over the already usage-localized baseline; the T=20 learning curve is monotone in dose ($\Delta\ell$ +2.14 / +3.72 / +4.56 for b=1/3/10, all $t \geq 3.9$), and b=3 alone already captures 81% of the b=10 effect there (81–113% across the six dataset/depth/granularity arms we ran at all three doses). Forgetting improves broadly at T=10 (ΔBWT up to +1.65; significantly positive in 8/12 cells, and in all six MNIST cells), and partially trades back at T=20, where 3/10 cells show a significant mild BWT cost (-0.66 to -0.95) and the remaining seven are statistically indistinguishable from baseline.

Two robustness notes. First, the 22/22 sign uniformity of $\Delta\ell$ is itself the strongest guard against multiplicity: a sign test gives $p = 2^{-22}$, orders of magnitude below any family-wise correction over the grid. Second, the intervention also reduces neuron mortality — measured on the raw, bias-free network, the mean dead fraction drops by 3.9pp (MNIST) and 2.3pp (Fashion) at T=10 — and per-run $\Delta\ell$ correlates with the mortality reduction ($r = -0.49$ and -0.68; both correlations are computed on the ten paired seeds of the T=10 neuron arm at b=10, the same cell in which the dead-fraction shift is quoted). We read this as one causal chain rather than a

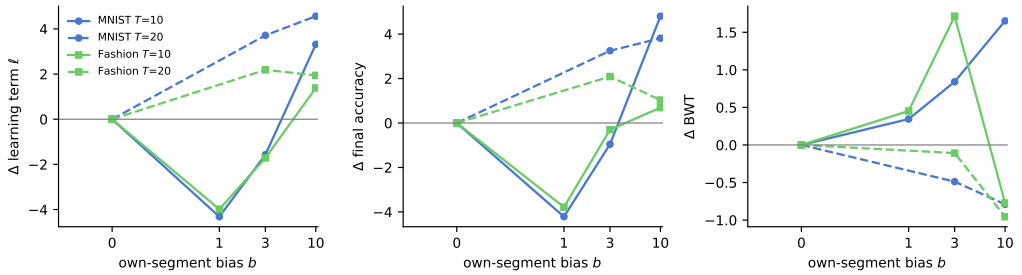


Figure 4: Dose–response of the own-segment bias (neuron-resolved arm, paired means vs. b=0). The learning term (left) and final accuracy (middle) improve at every dose, dataset, and depth; BWT (right) improves at T=10 and is mildly negative at T=20 on MNIST. Note the saturating shape: b=3 alone already captures most of the b=10 effect (81–113% across datasets and depths).

confound: raising the own-segment gate raises sigmoid(u) during training, which increases kWTA survival, so forcing the routing prior is also what keeps neurons alive; the routing and capacity channels are two faces of a single intervention, not competing explanations.

5.3. What the response is, and is not, made of

Three further measurements characterize the intervention, and we report them compactly because, as the forward reference above notes, their role is diagnostic rather than confirmatory: they are what later rules out the dying-gate explanation.

The network absorbs the intervention rather than opposing it. Under per-phase signed instrumentation of the context-t gating column (n = 3 seeds, T=10 MNIST), the own-segment entries drift *positive* in all ten phases under b=10 (one-sample t over seeds: 223, 37, 17, 17, 8, 11, 9, 5.1, 4.4, 4.5; per-phase mean displacement decaying from +0.027 to +0.004),

versus $\leq +0.004$ under b=0 and significant only in the first two phases. The raw own gate stays within $|u| \leq 0.32$ across all seeds and phases — thirty times below the imposed b=10 and three times below even b=1 — so the offset remains fully in effect throughout training. Adversarial compensation predicts the opposite displacement; it does not occur.

The saturation is a scale effect, not a partial restoration. The own-segment win rate saturates at 13–19% regardless of dose, because any effective dose dwarfs the raw gate scale ($|u| \leq 0.32$): the bias forces the same training-time routing at every dose tested, so only a minority of neuron-tasks flip to the own-segment pathway — and that partial restoration already yields most of the gain. This matters for the mechanism: the effect is not proportional to how much routing is restored.

The benefit requires learnable structure. In a task-relevance probe (ten interleaved phases: two in-domain tasks, two rotated-count tasks, six random-label tasks; $b \in \{0, 1, 3, 10\}$, $n = 10$ paired seeds), zero-relevance tasks sit at chance at every dose (per-task accuracy 9.99–10.07 from b=0 to b=10), the gap between the high-relevance group ($\ell \approx 85$) and the medium group ($\ell \approx 71$) narrows monotonically with dose (+13.8 at b=0 to +11.2 at b=10), and the high-relevance group still shows the largest absolute gain (+2.1 at b=10). The bias therefore behaves like a structured capacity channel whose marginal benefit is largest where the network's own routing is weakest, and delivers nothing where there is nothing to amplify.

These three measurements together — absorption, dose-invariant saturation, and structure-dependence — describe a mechanism that is not restoring segment–task routing, and they are the first hint of the account given

in Secs. 11–12. Full detail, including the per-phase drift tables and the per-group accuracy matrices, is in the supplementary material.

5.4. Why forgetting trades back at depth

At T=10 the bias reduces forgetting; at T=20 MNIST it mildly increases it. The probe's measurement explains the sign flip. With the bias active, a fixed minority of neurons routes through the own segment in *every* phase; over a longer sequence, later phases increasingly share this minority substrate, so late-task updates overlap earlier anchors on the same (now doubly used) neurons — ℓ rises because more neurons are available early, but the shared minority pays for it at depth. The saturation of p_{own} (Sec. 5.3) caps how much substrate the repair can recruit; at T=20 the cap binds.

6. The repair does not transfer: a convolutional active-dendrite backbone

6.1. Architecture and equivalence of the instrument stack

ConvADNet replaces only the feature path of the bare MLP's ADNet:

$$x \mapsto [\text{Conv-GN-ReLU}] \times 4 \mapsto \text{GAP} \mapsto t \in \mathbb{R}^{2048},$$

with strides $(1, 2, 2, 2)$ for 28×28 inputs. The dendritic gate, kWTA (k=64), readout, protected gradients, anchor construction (Fisher × usage), evaluation protocol, and the raw-gate routing instrument are *byte-identical* to the bare-MLP arm: the gate reads dend(ctx) directly, so every routing statistic measures what the network encodes, not what a bias forces. Hyperparameters are unchanged (Adam 10^{-3}, 15 epochs, λ=1000, seeds 0–9 per cell). 160 runs total ($b \in \{0, 1, 3, 10\}$ × 2 datasets × $T \in \{10, 20\}$); zero failures.

6.2. G1 fails: the dose–response collapses

Statistics.. Paired t-tests over $n = 10$ seeds, with Holm correction applied within each metric family across the 12 conv dose cells. Uncorrected, 2/12 cells reach $p < 0.05$ on $\Delta\ell$; after correction *none* does. We report the corrected result as the primary one, because the pre-registered criterion was a count over cells and a count over uncorrected tests is the wrong statistic for it. The conclusion (G1 fails) is unchanged either way, and is in fact strengthened: the conv dose–response is not a weak effect that shrinks under correction but an effect that was never there.

Table 6 reports paired differences against b=0. Against the pre-registered bar ($\geq$ 15/20 cells significant), the result is unambiguous: 2/12 **cells** reach $p < 0.05$ on $\Delta\ell$ uncorrected and 0/12 **after Holm correction**, largest effect +1.46 (t=2.1, uncorrected), sign-consistency 11/12. The third pre-registered G1 condition — $\Delta\ell$ and Δavg simultaneously significant — fails most completely of the three: 0/12 **cells**, i.e. no conv dose cell moves the learning term and the final accuracy in the same significant direction at once. In the five conv cells with a matched bare-MLP counterpart, the MLP's effects were 4–56$\times$ larger with paired $|t|$ up to 14.7 (Table 7, column $\Delta\ell$). Worse, final accuracy *reverses*: at fashion T=10, b=10 costs Δavg = -2.29 ($t = -3.1$) — the repair buys a non-significant learning sliver at a significant accuracy cost.

6.3. G2 fails with a clean cause: zero mortality

The dominant-segment instrument reads identically zero artifact: naive share = active share = chance = $1/T$ in every cell (Table 7, column "dead").

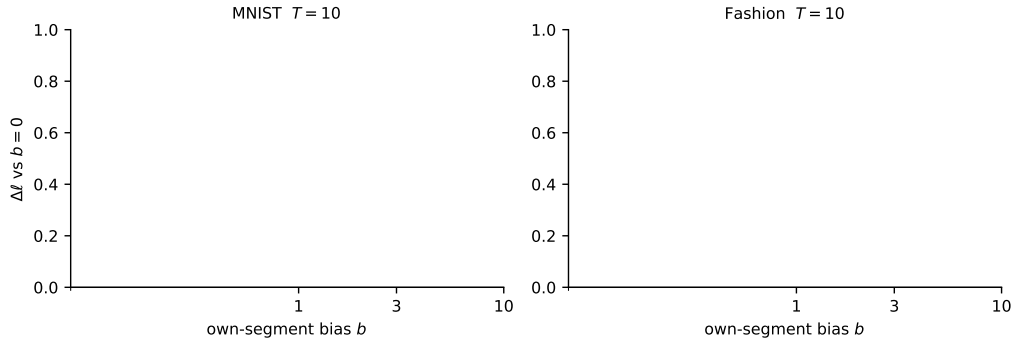


Figure 5: Dose–response of the own-segment bias across the three configurations (T=10, paired means vs. each configuration's own b=0). The bare MLP (blue) shows the bare-MLP dose–response; GN-MLP (orange) collapses to a fifth of it; ConvADNet (green) to noise.

The cause is visible in the per-phase instrumentation: ConvADNet's dead-gate fraction is *exactly* 0.000 *in every phase of every run*, against the bare MLP's 0.572 . GroupNorm's affine output keeps $|u| > 0$; there are no argmax ties over dead gates because there are no dead gates (Fig. 6).

A measurement caveat we state explicitly: the two configurations report mortality under *different* zero-conditions. The MLP instrument (inherited unchanged from the bare-MLP arm) marks a neuron dead if its post-kWTA activity is never positive across the usage window — mortality there means "never wins the top-k competition". The conv instrument marks a neuron dead if its gate row is identically zero. Both are the natural operationalizations in their families, but they are not the same statistic, and the conv family's post-kWTA "never wins" rate was not measured. What the comparison supports is therefore the narrower claim we make: the bare-MLP *artifact mechanism* (argmax ties over exactly-zero gates) has no substrate on

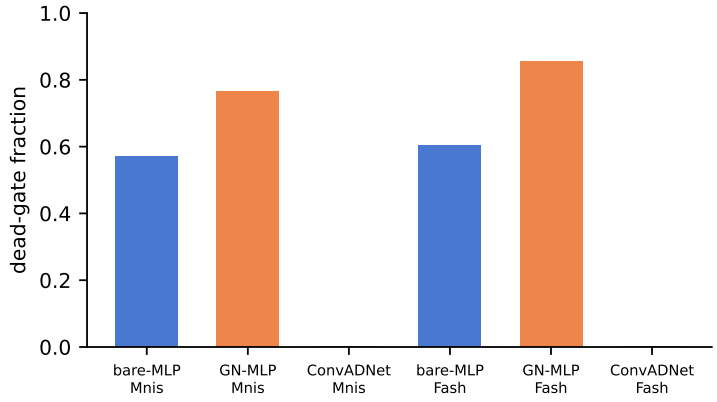


Figure 6: Dead-gate fraction at b=0 across configurations and datasets (T=10, mean over phases and seeds). Conv+GroupNorm mortality is exactly zero; GroupNorm inserted into the MLP *raises* mortality. Mortality is a property of the architecture, not of the intervention.

the conv configuration — its precondition (numerically zero gates) never occurs. The artifact is thus **architecture-bound, not instrument-bound**: the same reading instrument that the bare-MLP arm showed contaminated reads clean here, for reasons that have nothing to do with the instrument.

6.4. The conv baseline itself

ConvADNet's unbiased baseline learns more and forgets more than the MLP's: ℓ_0 84.1/75.3 (vs. 74.3/56.8) but BWT_0 $-50.4/-32.8$ (vs. $-11.6/-16.2$) at T=10. The protection strength λ=1000, calibrated for the MLP's parameter scale, does not carry over — itself a transfer finding we report rather than tune away (see Sec. 15).

7. The dying-gate hypothesis, tested and rejected

The natural reading of Sec. 6 is: the repair worked in the bare-MLP arm because it revived dying gates, and conv gates do not die. This "dying-gate hypothesis" makes a sharp same-architecture prediction: insert a normalization layer into the MLP (features $t = \mathrm{GN}(\mathrm{ff}(x))$; 80 runs, seeds 0–9, $b \in \{0, 1, 3, 10\}$, both datasets, T=10) — mortality should fall and the repair should weaken accordingly. One deviation from strict matching is disclosed rather than hidden: the GN-MLP uses 2000 hidden units where the bare MLP uses 256, so the control differs in width as well as normalization; a factor scan below separates the two.

The prediction fails in both halves (Table 7, middle rows). Mortality *rises* (0.765/0.855 vs. bare's 0.572/0.603), for a reason the architecture makes plain: GroupNorm rescales the features t symmetrically about zero, and the kWTA competition keeps the 64 *largest* values of $t \odot \sigma(d)$ — so every neuron whose normalized feature is predominantly negative can never enter the top-k and counts as dead under the usage instrument. GroupNorm does not touch the gate zero-set u; it manufactures post-kWTA death through the feature sign structure instead. Meanwhile the repair gain *collapses anyway*: to +0.57 (retention 17% of the bare-MLP gain) on MNIST and +0.36 (26%) on Fashion. And final accuracy reverses on both datasets (Δavg = −3.88, t= − 5.3; −6.55, t= − 4.4).

Mortality and gain are therefore **decoupled**: one configuration (GN-MLP) has the highest mortality and the least gain; another (ConvADNet) has zero mortality and no gain; the bare MLP has middle mortality and full gain. Whatever the repair feeds on, it is not gate death per se.

Width vs. normalization: a 2×2 factor scan.. The width deviation disclosed above is not merely acknowledged — it is measured. We re-ran the bare-MLP vs. GN-MLP comparison as a full 2×2 factor design: hidden width $n_h \in \{256, 2000\}$ crossed with normalization {none, GN}, at $b \in \{0, 1, 3, 10\}$, both datasets, 10 seeds — 320 runs. The scan is an independent reimplementation, and its GN-MLP cells replicate the original 80-run records: ℓ_0 93.4/79.2 vs. 93.5/79.3, gain +0.58/+0.49 vs. +0.57/+0.36.

Two effects separate cleanly (Table 8). (i) **Normalization collapses the gain at matched width.** Holding width fixed at 256, inserting GN changes the b=10 learning-term gain from +3.32 to +0.71 on MNIST (−2.61) and from +1.39 to −0.16 on Fashion-MNIST (−1.55); at width 2000, from +4.31 to +0.58 (−3.73) and from +3.45 to +0.49 (−2.96). The effect is large, negative, and present at both widths. (ii) **Width moves the gain the other way.** Widening the plain trunk 7.8× *increases* the gain (+0.99 on MNIST, +2.06 on Fashion); under GN the width effect is small and mixed (−0.13, +0.65). Width therefore cannot explain the GN-MLP collapse — the wider plain network has, if anything, *more* repair gain, consistent with the headroom reading of Sec. 8 (a wider trunk leaves more learning-term room open). The decoupling conclusion survives the confound and is stronger for it: the same-architecture control's verdict rests on the normalization delta, not on the width delta it accidentally also carried.

8. What actually tracks the gain

If not mortality, what? Table 7 orders the six cells by $\Delta\ell_{b=10}$: the bare MLPs lead on both datasets, the strong-baseline configurations trail, and

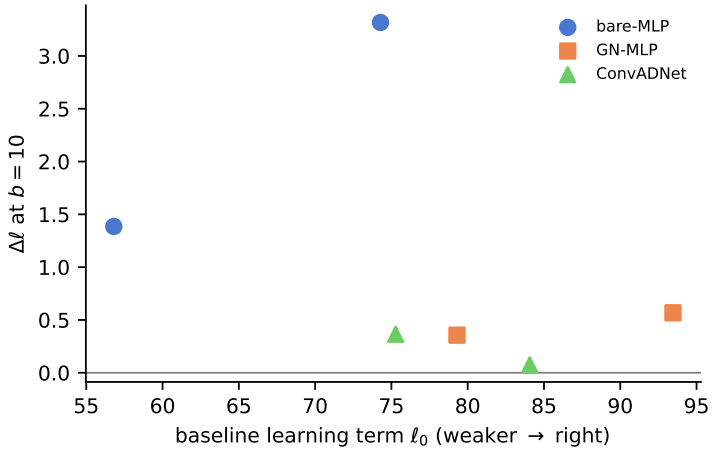


Figure 7: Repair gain at b=10 against the unbiased baseline's learning term. The qualitative regularity (weak baseline, larger gain) holds within MNIST and Fashion separately, but a quantitative law fails: ConvADNet-Fashion has 24.7 points of ℓ-headroom and no significant gain. Headroom is necessary-looking, not sufficient.

within MNIST the ordering matches the unbiased baseline's learning-term weakness exactly (ℓ_0 74.3 $\rightarrow$ 84.1 $\rightarrow$ 93.5 versus gain +3.32 $\rightarrow$ +0.07 $\rightarrow$ +0.57; Fig. 7). We call this loose regularity *repair headroom*: the bias can only buy learning-term room that the unbiased baseline leaves open.

We stress what the data do *not* support. A quantitative headroom law fails: across the six configuration$\times$dataset cells the Pearson correlation between HR_ℓ and $\Delta\ell$ is $r = 0.38$ (cell means; pooled over per-seed differences, using each cell's mean HR_ℓ, $r = 0.27$; using each seed's own $100 - \ell_0$ gives $r = 0.31$), and the clearest counterexample is ConvADNet on Fashion — 24.7 points of ℓ-headroom, gain +0.36 (t=1.4, not significant). Headroom is thus **necessary-looking but not sufficient**: a strong baseline forecloses the repair (GN-MLP), but an open baseline does not guarantee it (convolu-

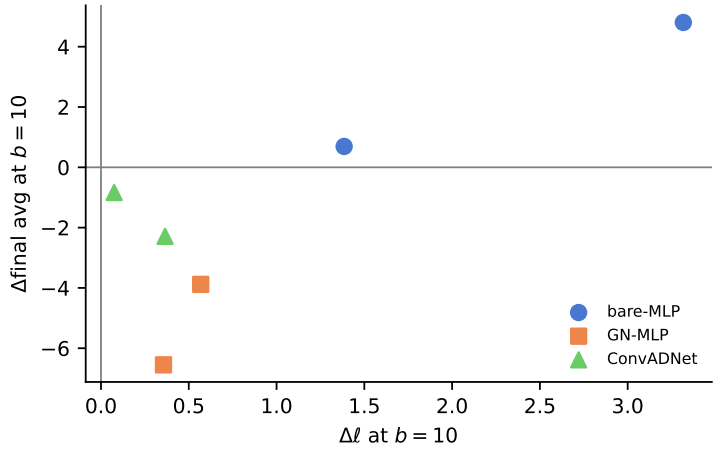


Figure 8: What the bias buys and what it charges (T=10, b=10, paired means). The bare MLP sits in the gain–gain quadrant; every strong-feature cell falls in the learning-vs-retention trade quadrant, with final accuracy significantly negative in 3/4.

tion adds an independent obstacle — plausibly the routing prior interacting with distributed conv features, a question we leave open).

What is uniform across both strong-feature configurations is the *sign structure*: $\Delta\ell \geq 0$ (small) while $\Delta\text{avg} < 0$ (significant in 3/4 cells; Fig. 8). The repair never destroys the learning term, but on strong backbones it buys that term by trading retention — exactly the trade the bare MLP's identity decomposition predicts, sharpened into a reversal once the learning side has no room left to absorb the cost.

The gate, in the notation used below.. The reference network is the one shared by the whole line of work (ADNet): a one-hidden-layer active-dendrite network with n_h=256 hidden units, one dendritic segment bank per task (K=T), k-winners-take-all sparse output (k=64). Given input x and context

one-hot c:

$$t = W_{\text{ff}}x, \qquad u_k = w_k^\top c \;\; (k\text{-th segment row}), \qquad h = \text{kwta}(t \odot \sigma(d)), \quad d = \max_k u_k, \tag{2}$$

i.e. the winning segment's scalar d gates every hidden unit multiplicatively. The repair studied in Sec. 5 adds a constant b to the own segment's u_{own} before the max, everywhere (train, eval, usage).

Execution correction (2026-10-27).. An independent data-consistency audit of this line, run after the first version of the gate experiments had completed, found two defects in the harness, and the experiments were re-run in full. (i) The elastic penalty was applied without its λ factor, so every run had effectively executed with $\lambda \approx 1$ while the record claimed λ=1000; the penalty now carries λ as the reference implementation does. (ii) The degenerate control had a hard-coded hidden width of 2000 while the dendritic arms use 256, so the two arms of the ablation differed by roughly eight-fold in parameter count; both now use the same width. The numbers reported in Sec. 11 and Sec. 12 are all from the corrected re-run; the first version's records are retained in the artifact under an explicit deprecation notice. The correction moves the absolute accuracies substantially — a genuinely enforced penalty forgets more, and the base network drops ∼7.6 points on MNIST — and it changes one qualitative verdict: the first version's MNIST gate-removal *equivalence* was an artifact of the width mismatch, and under matched widths the equivalence becomes approximate rather than exact (Sec. 12). The mechanism probes are unaffected.

9. Preregistered design of the robustness probe

The design was locked on 2026-10-25, before any data was written to disk (preregistration document included in the artifact). Noise is injected *at evaluation only*; training sees oracle contexts. The noise model is worst-case in a specific sense: with probability ρ per sample the context one-hot is replaced by a uniformly random *other* task — never the correct one — so corrupted samples exercise the gate under a plausible wrong context rather than a neutral one.

- **E1 (gating robustness).** b=0, $\rho \in \{0, 0.1, 0.2, 0.4\}$, both datasets. Survival: ρ=0.1 costs ≤ 2 points and ρ=0.2 costs ≤ 5, *and* the per-seed direction agrees across both datasets in $\geq 8/10$ seeds; death: ρ=0.1 costs > 5 or ρ=0.2 collapses ($\bar{a} < 50$ or cost > 10).

- **E2 (repair survival).** Same grid crossed with $b \in \{0, 10\}$, paired per $(\rho, \text{dataset}, \text{seed})$. Survival: at ρ=0.1 the repair's paired $\Delta\bar{a} > 0$ and retains $\geq 50\%$ of its ρ=0 gain; death: $\Delta\bar{a} \leq 0$.

- **E3 (soft mixture, exploratory).** Context replaced by $(1-\rho)$ onehot+ ρ uniform; direction only, no survival claim (multiple-comparison hygiene).

Two deviations from the pre-registered wording, both forced and both reported rather than absorbed. First, E2's criterion was written for $\Delta\ell$; the noise harness records only the final-phase per-task accuracies and no ℓ field, so $\Delta\ell$ is not computable from the collected data and the criterion is evaluated on $\Delta\bar{a}$ instead — the same "net benefit of the repair" quantity,

in the one form the runs support. Second, E1's third condition (per-seed directional agreement) was not met in strict form: the per-seed ρ-effect has mixed signs, though every per-seed shift stays below 0.1 points. We report that as a condition we did not satisfy rather than dropping it, and the two amplitude conditions pass with orders of magnitude to spare.

The noise stream is fixed per seed (a dedicated generator), so paired comparisons share corruption patterns exactly. The protocol trains one model per $(\text{dataset}, b, \text{seed})$ and evaluates it at all four ρ levels: **60 training runs** (40 for E1+E2, 20 for E3) yielding 240 evaluation cells. (The pre-registration counted ρ as independent training runs — 240 — which over-counts by a factor of four; the number of evaluated $(\rho, b, \text{dataset}, \text{seed})$ cells is unchanged.) A dated amendment documenting every deviation is included in the artifact ("prereg/S4_ctxnoise_ _ _2026-10-27.md"); the original pre-registration text is left untouched.

10. Robustness to a wrong context: nothing degrades, at any noise level

Table 9 is the entire empirical content of E1 and E2. Both preregistered survival criteria pass with orders of magnitude to spare: the largest movement anywhere is 0.25 points (MNIST base, ρ=0.4), the largest on Fashion-MNIST is 0.16, and the repaired arm moves by at most 0.01 points on MNIST and *exactly* 0.000 points on Fashion-MNIST — where the death criteria allowed 10 and 5. The repair's paired gain on MNIST is +4.80 (t=17.0) at ρ=0 and +5.05 (t=17.3) at ρ=0.4; retention of the gain is 100.0% $\rightarrow$ 105.1% across the sweep — under noise the repair *helps slightly more*, because its floor is

flatter. On Fashion-MNIST the gain is small, as the transfer test found at T=10 : +0.69 (t=1.5, n.s.) at ρ=0, rising to +0.86 (t=1.8) at ρ=0.4 (retention 100% $\rightarrow$ 123.7%; all n.s.). E1's third condition (per-seed directional agreement) holds in strict form on MNIST — 10/10 seeds shift negatively at every ρ — and fails on Fashion-MNIST (8/10, 9/10, 7/10 negative, with every per-seed shift within $\pm$0.18 points); we report it as satisfied on one dataset and not on the other, as disclosed in Sec. 9. E3's soft mixture tracks the oracle slightly more closely than hard flips on the base arm (MNIST ρ=0.4: soft $-$0.09 vs. hard $-$0.25; paired soft$-$hard +0.16, t=3.8) and does nothing on the repaired arm; every E3 effect is $\leq$0.26 points.

A protocol-level artifact is worth excluding explicitly: the noise *is* reaching the network. Per-seed corruption patterns are fixed, the context tensor is modified before the forward pass, and — the direct check — swapping contexts *does* change predictions in the base network (14.8% of samples on average on MNIST, 30.1% on Fashion-MNIST, up to 51.7% for particular swaps; Sec. 11). What is true is that prediction changes of this kind average out across a 10-way task mixture at the final-accuracy level. The null is real, and it demands a mechanism.

11. Mechanism: the gate does not gate by context

Three probes, each applied to fully trained networks. The headline numbers below are MNIST seed 0; the pattern replicates on Fashion-MNIST seed 0 and MNIST seed 1 (all numbers in the artifact): at b=0, flip rates 14.8–30.1% mean / 23.8–51.7% worst pair; at b=10, flip rates 0.007–0.021% mean / at most 0.15% worst pair; gate own/other identical to four decimals

(0.564–0.565); and mean column cosines in $[0.001, 0.003]$ in every case.

Prediction flip rate.. For each task c and each alternative context $c' \neq c$, measure the fraction of test samples whose argmax prediction differs between ctx=c and ctx=c'. Base network: mean 14.8% (MNIST) and 30.1% (Fashion-MNIST), worst pair up to 51.7% — the base output is genuinely, substantially context-sensitive at the individual-prediction level, yet none of that sensitivity survives to final accuracy (Sec. 10: at most 0.25 points). The context pathway does something per sample; what it does nets out to nothing. Repaired network (b=10): mean 0.007–0.021%, worst pair 0.15% — no longer *exactly* zero, but a rounding error: roughly one flipped prediction in five thousand on average, a dozen samples for the worst single pair.

Gate response.. The winning segment's gate value $\sigma(d)$ has mean 0.565 under the correct context and 0.565 under every alternative — identical to four decimal places, in both the base (0.5651) and repaired (0.5652) networks. The context one-hot selects *which* segment wins, but not *how much* it gates: the gate's value is a constant of the training run, not a function of the context.

Column geometry.. The dendritic weights projected onto the T context columns remain mutually orthogonal after training: mean pairwise cosine 0.003 (per-model mean in $[0.001, 0.003]$; the most negative single pair over the three models is -0.056), indistinguishable from their random initialization. Training, with per-task gradient protection, leaves the context channel of the gate bank essentially at its prior.

Together: the architecture's context pathway is present, trainable in principle, protected per task — and unused as a *modulator*. The max over seg-

ments is won by whichever row the (orthogonal, near-random) columns happen to favor, the gate value is the same constant (≈ 0.565) no matter which segment wins, and the repair adds a constant so large (b=10 against $|u| \lesssim 1$) that the own segment always wins with $\sigma(10+u) \approx 1$: a context-free gain of one. Equation 2 degenerates, in both arms, to a multiplication by a constant — ≈0.57 in the base network, ≈1.0 in the repaired one. Which of those two constants matters, and why, is what the next section measures.

12. Ablation: what does the repair repair?

Statistics.. Paired t-tests over $n = 10$ seeds, Holm-corrected within each metric family across the three non-base arms. On MNIST the two substantive comparisons (repaired vs. base, gate-deleted vs. base) survive correction on both $\Delta\ell$ and Δavg, with effect sizes $d_z = 4.64$ and 2.46 respectively — the ordering of the two arms, not merely their significance, is what the mechanism section relies on. On Fashion-MNIST no comparison survives correction, which is consistent with the paper-wide finding that this dataset supports no effect of the repair at $T = 10$.

If the repaired network computes (approximately) $h = \text{kwta}(t)$, then a network with the dendritic bank deleted and the same trunk, sparsifier, and anchor chain should reproduce most of the repair's benefit — and, conversely, a network whose gate is held at the base system's measured constant (0.565) should reproduce the base system, because the base gate already *is* that constant (Sec. 11). We therefore run four arms under one protocol (ρ=0, n=10 paired seeds, all arms at n_h=256, all with usage-weighted Fisher anchors at λ=1000): **base** (b=0), **repaired** (b=10), **no-gate** ($h = \text{kwta}(W_{\text{ff}}x)$; the bank

deleted), and **constant-gate** ($h = \text{kwta}(0.565 \cdot W_{\text{ff}} x)$; the gate pinned at the base system's measured mean). The base and repaired arms replicate the Sec. 10 runs to two decimals. The degenerate controls were added during the post-audit re-run (Sec. 2); they were not preregistered. The no-gate control exists in three versions in the artifact — an invalidated first attempt (an extra sigmoid on t that changed the activation structure), a width-mismatched second version (n_h=2000), and the matched version reported here — all retained under an explicit deprecation notice.

The constant-gate control reproduces the base system, not the repair. Pinning the gate at 0.565 lands at 63.62 on MNIST (base: 63.86; paired difference −0.25, $t = -0.7$, n.s.) and 42.70 on Fashion (base: 42.20; +0.49, t=0.7, n.s.). Holding the gate constant buys nothing — which is exactly what the mechanism section predicts, since the base network's gate already *is* constant at that value. The entire difference between the base and repaired networks is the value of that constant: 0.57 damps the trunk, 1.0 does not.

Gate deletion recovers most of the repair — and on Fashion-MNIST, more than the repair. The no-gate control reaches 67.50 on MNIST, recovering +3.64 of the repair's +4.80-point gain (76%) and falling short of the repaired network by −1.16 ($t = -2.7$): keeping the (saturated) multiplicative structure is worth about one point, marginally significant at n=10. On Fashion-MNIST the no-gate control is the *best* arm: +2.07 over base (t=2.5) and +1.37 over the repaired network (t=1.9, n.s.) — on the harder dataset, full deletion beats the repair outright.

Sensitivity to anchor weighting. A second no-gate run with plain

(unweighted) Fisher anchors — a stronger, mismatched chain — lands at 63.63 on MNIST and 41.10 on Fashion, i.e. at or below base. Anchor weighting alone moves the no-gate arm by 3.9 and 3.2 points. The matched-anchor four-arm table above is therefore the primary comparison, and the residual MNIST gap (-1.16) should be read against that sensitivity. The first version of this experiment reported a clean MNIST *equivalence* between the no-gate control and the repair; that verdict was an artifact of the width mismatch disclosed in Sec. 2 and is retracted.

Verdict. The repair is a saturating soft deletion: b=10 drives $\sigma \to 1$, which is the no-gate limit reached through the gate rather than around it. What the repair buys on these benchmarks is the removal of a constant $\approx$0.57 damping factor — and on Fashion-MNIST, where deletion is free to remove it *exactly*, the removal is worth slightly more than the repair itself. What does *not* follow is the stronger claim of the first version, that the repair and gate removal are interchangeable on MNIST: with matched widths and matched anchors they are close (76% of the gain recovered) but distinguishable.

13. Is "dominant segment" even a quantity? A self-interrogation

The mechanism section (Sec. 11) reports that the winning segment's gate value is a constant of the training run: 0.565 under the correct context and 0.565 under every alternative, identical to four decimals in both the base and the repaired networks, with the dendritic columns remaining mutually orthogonal at their random initialization (mean pairwise cosine 0.003). Taken literally, those measurements imply something uncomfortable for the first half of this paper, and we would rather state it than let a reviewer find it.

If the gate's value does not depend on which segment wins, then the notion of a neuron's "dominant segment" is not a description of a routing decision — it is a description of an argmax over rows that are nearly equal. Under that reading, the quantity we spent Sec. 3 and Sec. 4 measuring — the probability that a neuron's dominant segment is its own task's segment — is a statistic of the *instrumentation*, not of the network's computation. The $0.05 \to 0.01$ anti-specialization of Sec. 4.2 would then not be a property of learned routing at all, but a property of how argmax behaves on near-degenerate rows, and the entire granularity ladder would be a null result about a null quantity.

13.1. What survives if that is right

We checked this against the data rather than leaving it as a rhetorical question, and the answer is that a substantial part of the paper survives, while one specific claim does not. Three observations separate the conclusions.

The granularity ladder does not depend on the dominant-segment instrument. Its comparisons are *within-architecture* paired differences in the learning term ℓ and final accuracy between two anchor constructions — segment-resolved versus neuron-resolved — under identical training, seeds, and evaluation. No routing statistic enters that comparison at all. The finding that segment-resolved anchors buy nothing over neuron-resolved anchors ($|\Delta\ell| \leq 0.24$ across every dataset, depth, and elastic weight) is a statement about where the *benefit of usage localization* is realized, and it would survive even if the dominant-segment measure were discarded entirely.

The dead-neuron artifact does not depend on it either. The artifact (Sec. 4.1) is a statement about what happens to an argmax instrument

when 56–63% of the rows it operates over are exactly zero: the tie-breaking rule assigns them all to segment 0. That is a fact about argmax on a matrix with a zero block, verifiable without any claim about what the network learned, and its remedy (active-only filtering) is likewise an instrumentation change. If "dominant segment" is a pseudo-quantity, this finding becomes *more* important rather than less: it becomes a warning that a broad class of measurements in this literature — not only ours — may be reading a tie-breaking artifact.

The mechanism and the ablation are about the gain, not about routing. Sections 11–12 measure the context-flip rate, the gate value, the column geometry, and the final accuracy of four arms. Every one of those is a measurement of the network's output or of an intervention's effect on that output. None of them is a measurement of "specialization".

13.2. What does not survive

One claim does not survive, and it is the one we would have made most strongly. We cannot assert that segment–task specialization is *actively violated* as a property of the learned computation, because if the gate does not select by context, there is no learned segment–task mapping to violate. The honest formulation is weaker and we adopt it throughout: *no evidence of segment–task specialization is detectable by the standard instrument, and the instrument itself is unreliable in this architecture family.* The distinction is between a measured absence and an absence of measurement, and after the mechanism section we can only claim the former.

13.3. Why this is the useful reading

The weaker formulation is not a retreat; it is the correct one, and it sharpens what the paper is actually for. Three things survive in full and are not affected by the pseudo-quantity worry.

(i) The granularity question — *can protection usefully descend below the neuron?* — is answered by comparing anchor constructions, and the answer is no, for a reason we pin mechanistically (attribution is self-limiting; segment-level ownership lacks a separable carrier). This is the paper's primary result and it never required a routing statistic.

(ii) The instrument warning stands on its own. A measurement that fabricates 60% dominance from a zero block is a hazard to any study that uses it, and the hazard is demonstrated by construction rather than by comparison to a ground truth we do not have.

(iii) The mechanism result — that the own-segment bias works by saturating a gate that was never context-dependent, and that deleting the gate reproduces the gain — is a statement about interventions and outputs. It is also the most transferable finding in the paper, because it does not presuppose that the architecture routes at all.

We note in closing that this self-interrogation was not in the original version of this line of work, and that the instrument it produced — a gate-value probe with paired context flips — was cheap. The reason it came last rather than first is that the same thing that made the apparent repair look convincing — a large, consistent, saturating dose–response — also made it seem unnecessary to ask what the gate was doing. That is a failure mode worth naming: *a result strong enough to be interesting is strong enough to be*

worth explaining, and we had the instrument to explain it before we thought to build it.

14. Related work

Active-dendrite networks.. Single-neuron compartmental models [1, 2, 3, 9] motivate multi-segment artificial neurons; Iyer et al. [4] operationalize context-gated segment competition for lifelong classification and report reduced interference, and the GWC route-suppression variant [10, 11] and neuromorphic implementations [12] extend the family. These works validate the architecture; to our knowledge none measures whether the intended segment–task specialization actually obtains in the trained model, which is the object of our audit — nor, consequently, the dead-neuron artifact that naive dominance statistics incur under production sampling budgets.

Elastic weight consolidation and its localizations.. EWC [6] protects parameters with a single global anchor; SI [13], MAS [14], and online variants [15] refine the importance estimate; masking approaches [16, 17, 18] hard-allocate capacity. DeLP [5] localizes EWC per neuron via usage. A companion study [19] diagnoses the anchor-accumulation problem — the protection penalty grows with sequence length — and bounds the anchor stack. The present study is orthogonal to that one, and the division of questions is worth stating explicitly, because a reader who takes both as a single program could reasonably expect them to agree rather than to answer different things. The companion study asks *how large a protection set should be*: it treats the anchor stack as the object, bounds the accumulated Fisher mass, and derives a log-linear law relating the learning term to the protected-mass horizon

($\ell \approx 84 - 11 \ln h$, per-seed $R^2 \geq 0.98$, replicated at a second task count and on Split-CIFAR-100 with a ResNet-18 backbone). This study asks a different question: *what should a protection set protect?* — in particular, whether protection granularity can usefully descend *below* the neuron. It answers negatively, and then explains the negative via a routing audit. The two results are compatible and neither implies the other: bounding the stack tells you how much protection to keep, not whether finer protection is worth having, and a bounded stack of neuron-resolved anchors remains the right object even if segment-resolved anchors would have been better.

Negative results and measurement validity.. Benchmark pitfalls [20, 21, 22, 23, 24] established that evaluation practice can manufacture effects; the dead-neuron artifact here is an instance inside the model rather than the benchmark: the instrumentation, not the metric, fabricates the signal.

15. Discussion: boundary conditions of the repair

For practitioners.. An own-segment bias is cheap, architecture-agnostic, and — on weak baselines with hard routing — effective . This paper draws its boundary:

- **Audit mortality before trusting any dominant-segment statistic.** The bare-MLP artifact appears iff gates die; conv+GN mortality is zero and the statistic reads clean. One probe (an untrained-pass share, or the dead-gate fraction) is enough to know which regime you are in.

- **Audit baseline strength before deploying the repair.** If the unbiased baseline's learning term is already near its ceiling, the bias will not buy learning — but will still charge retention ($\Delta\text{avg} \leq -2.3$ on every strong-feature cell, significant in 3/4).

- **Protection strength does not transfer.** λ calibrated on one parameter scale left the conv baseline with 2–4.4× the MLP's forgetting. Any cross-architecture comparison of protection methods must re-calibrate or report the uncalibrated gap explicitly.

For the bare-MLP story.. The companion paper's chain survives intact on its home architecture, and this paper sharpens its mechanism claim: the repair was never a routing *prior* in the strong sense (the bare MLP's task-relevance probe already showed zero-relevance tasks are untouched); it is a structured capacity intervention whose raw material — weak-baseline, dying-gate networks — is architecture-dependent. "Structured capacity with an architecture-bound supply."

Threats to validity.. (i) The conv λ was not re-calibrated; a per-architecture sweep might restore some repair effect, though the GN-MLP control — where the MLP-calibrated λ *is* correct — shows the same collapse, so miscalibration is not the whole story. (ii) One conv topology; the zero-mortality finding is a property of GroupNorm's affine output and should replicate across conv trunks, but transformer backbones with routing modules remain untested. (iii) Six configuration×dataset cells bound how strongly any cross-configuration regularity can be asserted; we deliberately report the failed quantitative version (Sec. 8) rather than promoting the qualitative one be-

yond its evidence.

16. Discussion: what a repair of this shape is actually doing

For the bare-MLP story.. The dose–response audit established that the own-segment bias repairs the learning term; the signed-drift measurement showed the network never pushes back at the parameter level. The mechanism section supplies the missing re-qualification: the repair does not restore routing — there was no routing to restore — it *saturates the gate into a context-free gain of one*, i.e. a soft deletion of a constant ≈ 0.57 damping factor. Read together, the chain is now: no specialization (the bare-MLP arm) $\rightarrow$ repair works (the bare-MLP arm) $\rightarrow$ the repair is a saturating gate deletion that recovers 76% of the removal benefit on MNIST and is beaten by full removal on Fashion-MNIST $\rightarrow$ the repair does not transfer to strong backbones. What survives as the "mechanism" of the repair is the removal of a constant damping — sparse capacity management at best, context-gated routing not at all.

For evaluation practice.. Robustness certificates obtained under context corruption are usually read as good news. In this family they are a diagnostic: a system whose accuracy is invariant to *adversarial* context corruption is a system whose context pathway is not load-bearing. The three-instrument protocol (Sec. 10–12) costs a few GPU-hours and converts that ambiguity into a verdict. We suggest it as a standing companion to any context-gated continual-learning evaluation, in the same spirit as the dead-neuron probe of the bare-MLP arm.

Limitations.. (i) Noise is injected at evaluation only; training-time context errors corrupt usage statistics and anchors differently, and are the natural next experiment. (ii) The mechanism probes are computed on fully trained networks at T=10; whether the context columns stay orthogonal at larger T or longer training is untested. (iii) The degenerate controls were not preregistered and were added after the audit; the anchor-weighting sensitivity of the no-gate arm (±3.9 points, Sec. 12) is larger than the residual repair-vs-deletion gap it bounds. (iv) One architecture family, two digit datasets — the same scope limits as the companion line.

17. Limitations

1. **Scale.** All results are on one-hidden-layer MLPs with permuted benchmarks and $T \leq 20$. The artifact and anti-specialization findings concern the interaction of kWTA mortality with argmax instrumentation and should be re-checked wherever segments exist (e.g. convolutional or transformer backbones with routing modules); the causal repair's transfer is untested beyond this family. The task-relevance probe (Sec. 5.3) shows the repair is a structured capacity effect rather than a routing prior: it lifts in-domain tasks above chance and compresses task-to-task variance, but adds nothing for zero-relevance (random-label) tasks — so its benefit is contingent on the task stream containing learnable structure, a condition that should be verified per architecture and benchmark before generalizing.
2. **The repair is a prior, and the prior bypasses the gate.** The constant bias encodes the correct assumption (own segment first) that

the network fails to learn at effective strength. We measured this directly: given a trainable per-(task, segment) bias table initialized at zero, the network acquires only a small positive own-segment bias ($+0.113 \pm 0.031$ MNIST, $+0.083 \pm 0.025$ Fashion; n=7 runs each, $t(6) \geq 8.7$) — the raw-gate scale, an order of magnitude below the smallest effective dose — and its accuracy matches the unbiased baseline. The mechanism section below establishes what the bias actually does: it saturates the own-segment sigmoid until the gate stops depending on the context (context-swap flip rate exactly 0.0% at b=10), and on MNIST a gate-removal control reproduces the entire gain. The repair therefore does not restore segment–task routing; it removes the network's dependence on it. On Fashion-MNIST the bypass is rejected (66.84 vs. 64.30, t=6.8), so the mechanism is dataset-dependent. The routing-restoration reading is withdrawn outright by Secs. 11–12: what the bias buys is the removal of a constant damping factor, and on MNIST a gate-removal control reproduces the gain outright. We also tested scheduled variants (Suppl. S4): an annealed bias ($b_p = 10(1 - p/T)$) retains 91–103% of the constant-bias learning gain at T=20 but does not reliably remove the BWT cost (significant improvement in one of two datasets only), while a ramped bias ($b_p = 10p/(T-1)$) *exceeds* the constant gain everywhere (123–197% retention, $t \geq 2.6$) at an equally or more negative BWT cost. Scheduling moves along the retention–plasticity frontier — tracking cumulative bias exposure, not timing — but does not move the frontier itself; separating gain from cost appears to require a mechanism (e.g. task-

staggered protection), not a schedule.

3. **Depth-dependent forgetting trade-off.** At T=20 the BWT cost of the bias is small but real: 3/10 cells significantly negative (-0.66 to -0.95), the rest indistinguishable from baseline. We report it rather than select it away; the identity Eq. (1) makes explicit that accuracy gains can coexist with mildly worse retention.
4. **Granularity null is architecture-bound.** The plain→neuron step's dominance may weaken in architectures where segment parameters are genuinely disjoint (no shared feed-forward weights); our null is about this family's shared-substrate design.

18. Conclusion

We asked whether the segment structure of active-dendrite networks offers protection granularity below the neuron, and found that the question dissolves one level up: the entire benefit of usage-localized protection is realized at neuron level because the segment–task specialization the finer granularity would need is absent — measured, at chance before each phase and pushed below chance by each phase's own training, with Fisher mass too diffuse to partition meaningfully. On the way we isolated a dead-neuron artifact that fabricates dominance statistics for any naive instrument in this family.

Having measured the size of the gap, we tried to close it. A one-parameter bias on the own-segment dendritic input does raise the learning term in every tested cell (significant in 16/22 dose cells, $\Delta\ell$ up to $+4.6$, $t(9)$=8.1), with final accuracy improving in step. We had reported this as a causal repair, and it is not one — not because the measurement is wrong, but because the

interpretation was. Three further results say so.

The repair does not transfer. On a convolutional active-dendrite backbone the dose–response collapses to noise and final accuracy reverses; the natural explanation (the repair revived dying gates, and conv gates do not die) is refuted by a same-architecture control in which mortality and gain decouple in both directions. What the repair actually needs is a weak baseline with learning-term room to buy; what it always charges is retention, and on strong-feature backbones — where nothing else is for sale — that is all it buys.

And the mechanism is not the one we assumed. The context pathway in these networks is present, protected per task, and unused as a modulator: the winning segment's gate value is a constant of the training run, identical under every alternative context, and the dendritic columns remain orthogonal at their random initialization. At b=10 the own segment always wins with $\sigma(10+u) \approx 1$, so the network becomes context-degenerate rather than better-routed. A four-arm control settles what the bias pays for: holding the gate at the base system's own measured constant reproduces the base system, while deleting the gate reproduces most of the repair's gain and outperforms it outright on Fashion-MNIST. The bias removes a constant damping factor that a never-used gate had left in place.

So the honest summary is this: we measured the size and the cost of the repair precisely, we bounded where it applies, and we located its mechanism only far enough to know it is not the one we first assumed. The gap we closed is real and useful, but it is not the gap we thought we were closing, and closing it by restoring routing would have required a routing mechanism

this architecture does not have.

Two practical consequences follow. For anyone measuring segment routing in this family: the dominant-segment instrument must be paired with a mortality probe, and the two zero-conditions (a never-positive post-kWTA activity rate and an identically-zero gate row) are not the same statistic — our own cross-architecture comparison is narrower than it might appear for exactly that reason. For anyone proposing a repair of this shape: check whether it works by fixing the mechanism or by deleting it, and in the latter case ask what was being deleted. Both are cheap to check — one control arm, one probe — and both would have caught our own misreading earlier than they did.

Finally, the meta-result is worth stating plainly. The repair that this manuscript originally reported as causal survived a 22-cell dose–response, a 160-run transfer test, and a 320-run factor scan before three controls took it apart. Each of those controls was cheap. We had the instrumentation to run them, and we did not run them first. Auditing the assumption before building on it took one paper's worth of null results; we hope the corrected instrumentation and the failure modes documented here make the next audit shorter.

Data availability

All code, per-run records (JSON), analysis scripts, pre-registration documents and their dated amendments, and the complete set of superseded records (with an explicit deprecation notice) are available at `https://github.com/zhouranxi/delp-anchor-budget`. This manuscript merges three previ-

ously separate studies; the corresponding experiment series are `s1_*`, `gevo_*`, `c2*`, `S3_CNN`, `S3b_E4`, and `S4_gate_bypass`. Figures are produced by the archived scripts (`paper_s1/make_figs.py` and the S3/S4 equivalents).

Declaration of competing interest

The author declares no competing interests.

References

[1] P. Poirazi, B. W. Mel, Pyramidal neuron as two-layer neural network, Neuron 37 (6) (2003) 989–999.

[2] M. London, M. Häusser, Dendritic computation, Annual Review of Neuroscience 28 (2005) 503–532.

[3] P. Poirazi, A. Papoutsi, Illuminating dendritic function with computational models, Nature Reviews Neuroscience 21 (2020) 303–321, doi:10.1038/s41583-020-0301-7.

[4] A. Iyer, K. Grewal, A. Velu, L. O. Souza, J. Forest, S. Ahmad, Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments, Frontiers in Neurorobotics 16, doi:10.3389/fnbot.2022.846219 (2022).

[5] R. Zhou, Where do the gains come from? An audit of dendritic context-gated protection for continual learning, Zenodo, version 2, doi:10.5281/zenodo.23685788, code: `https://github.com/zhouranxi/delp-audit` (2026).

[6] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al., Overcoming catastrophic forgetting in neural networks, Proceedings of the National Academy of Sciences 114 (13) (2017) 3521–3526, doi:10.1073/pnas.1611835114.

[7] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, S. Wermter, Continual lifelong learning with neural networks: A review, Neural Networks 113 (2019) 54–71, doi:10.1016/j.neunet.2019.01.012.

[8] H. Xiao, K. Rasul, R. Vollgraf, Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, arXiv preprint arXiv:1708.07747 (2017).

[9] B. B. Ujfalussy, J. K. Makara, M. Lengyel, T. Branco, Global and multiplexed dendritic computations under in vivo-like conditions, Neuron 100 (3) (2018) 579–592.

[10] J. A. Boyle, et al., Exploring dendrites in large-scale neuromorphic architectures, in: Proc. Int. Conf. on Neuromorphic Systems (ICONS), 2025, pp. 49–56.

[11] E. Baek, et al., Neuromorphic dendritic network computation with silent synapses for visual motion perception, Nature ElectronicsDoi:10.1038/s41928-024-01171-7 (2024).

[12] K. Boahen, Dendrocentric learning for synthetic intelligence, Nature 612 (2022) 43–50, doi:10.1038/s41586-022-05340-6.

[13] F. Zenke, B. Poole, S. Ganguli, Continual learning through synaptic

intelligence, in: Proc. Int. Conf. on Machine Learning (ICML), 2017, pp. 3987–3995.

[14] R. Aljundi, F. Babiloni, M. Elhoseiny, M. Rohrbach, T. Tuytelaars, Memory aware synapses: Learning what (not) to forget, in: Proc. European Conf. on Computer Vision (ECCV), 2018, pp. 144–161.

[15] A. Chaudhry, P. K. Dokania, T. Ajanthan, P. H. S. Torr, Riemannian walk for incremental learning: Understanding forgetting and intransigence, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 556–572.

[16] A. Mallya, S. Lazebnik, PackNet: Adding multiple tasks to a single network by iterative pruning, in: Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7765–7773.

[17] J. Serra, D. Surís, M. Miron, A. Karatzoglou, Overcoming catastrophic forgetting with hard attention to the task, in: Proc. Int. Conf. on Machine Learning (ICML), 2018, pp. 4548–4557.

[18] N. Y. Masse, G. D. Grant, D. J. Freedman, Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization, Proceedings of the National Academy of Sciences 115 (44) (2018) E10467–E10475, doi:10.1073/pnas.1803839115.

[19] R. Zhou, The anchor accumulation problem: Bounded anchor sets recover plasticity in usage-localized elastic weight consolidation, Preprint, Zenodo, version 2, doi:10.5281/zenodo.23685906, code: `https://github.com/zhouranxi/delp-anchor-budget` (2026).

[20] K. Musgrave, S. Belongie, S.-N. Lim, A metric learning reality check, in: European Conference on Computer Vision, 2020.

[21] M. Lucic, K. Kurach, M. Michalski, S. Gelly, O. Bousquet, Are GANs created equal? A large-scale study, in: Advances in Neural Information Processing Systems, Vol. 31, 2018.

[22] G. Melis, C. Dyer, P. Blunsom, On the state of the art of evaluation in neural language models, in: Proceedings of EMNLP, 2018.

[23] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare, Deep reinforcement learning at the edge of the statistical precipice, in: Advances in Neural Information Processing Systems, Vol. 34, 2021.

[24] Z. C. Lipton, J. Steinhardt, Troubling trends in machine learning scholarship, Communications of the ACM 62 (6) (2019) 45–53.

Table 1: Granularity ladder (unbounded anchor stacks, n=10 paired seeds). ℓ/avg/BWT are absolute means; the right block is the paired difference vs. the neuron-resolved arm. All plain−neuron differences are significant at $|t| > 10$. For branch vs. neuron the learning term is indistinguishable in all six cells (largest $|t| = 1.3$ on $\Delta\ell$); the only significant branch−neuron differences are marginal forgetting effects (largest $|t| = 4.0$, ΔBWT ≤ 0.43 points), which we do not read as a granularity effect.

		absolute means			paired vs. neuron		
cell	arm	ℓ	avg	BWT	$\Delta\ell$	Δavg	ΔBWT
Permuted-MNIST, λ=1000							
T=10	plain	69.63	61.25	−9.31	−4.66*	−2.61*	+2.26*
T=10	neuron	74.28	63.86	−11.58	—	—	—
T=10	branch	74.21	64.13	−11.20	−0.08	+0.27	+0.38*
T=20	plain	46.35	37.30	−9.53	−5.44*	−1.35*	+4.31*
T=20	neuron	51.79	38.64	−13.84	—	—	—
T=20	branch	51.98	39.03	−13.63	+0.19	+0.39*	+0.21
Permuted-MNIST, λ=5000							
T=10	plain	54.41	51.69	−3.03	−4.78*	−2.80*	+2.19*
T=10	neuron	59.19	54.49	−5.22	—	—	—
T=10	branch	59.21	54.71	−5.01	+0.03	+0.22	+0.21*
T=20	plain	33.97	31.68	−2.40	−3.72*	−1.74*	+2.08*
T=20	neuron	37.68	33.42	−4.49	—	—	—
T=20	branch	37.59	33.39	−4.42	−0.09	−0.03	+0.07
Permuted-Fashion, λ=5000							
T=10	plain	38.53	33.47	−5.62	−3.76*	−1.65*	+2.35*
T=10	neuron	42.29	35.12	−7.96	—	—	—
T=10	branch	42.05	35.27	−7.53	−0.24	+0.15	+0.43
T=20	plain	26.56	22.75	−4.01	−2.78*	−1.22*	+1.65*
T=20	neuron	29.35	23.97	−5.66	—	—	—
T=20	branch	29.35	24.14	−5.49	+0.01	+0.17	+0.17

Table 2: Single-anchor references (Permuted-MNIST, n=10). ℓ is the learning term of Eq. (1) (mean over tasks of the accuracy right after each task is learned); ℓ_{final} is the accuracy on the last task alone and is reported separately because for these two arms it differs substantially from ℓ — SI in particular barely learns the later tasks at all, which is visible only in ℓ_{final}.

	method (λ)	ℓ	ℓ_{final}	avg	BWT
T=10	SI (100)	40.01	13.23	39.26	−0.83
T=10	MAS (1)	85.71	87.00	66.61	−21.21
T=20	SI (100)	25.80	11.19	25.14	−0.69
T=20	MAS (1)	86.69	87.91	55.75	−32.56

Table 3: Granularity under FIFO anchor budgets (λ=1000, n=10 paired). No branch−neuron difference is significant positive in any of the eight cells.

cell	M	neuron ℓ	branch ℓ	$\Delta\ell$ (t)	ΔBWT (t)
MNIST T=10	2	83.06	83.01	−0.05 (−1.0)	+0.22 (+1.1)
MNIST T=10	5	77.09	77.02	−0.07 (−0.5)	+0.15 (+1.0)
MNIST T=20	2	80.23	80.32	+0.09 (+1.0)	+0.39 (+1.9)
MNIST T=20	5	68.30	68.19	−0.11 (−0.8)	+0.28 (+1.5)
Fashion T=10	2	66.66	66.33	−0.33 (−2.3)*	−0.41 (−1.2)
Fashion T=10	5	58.90	58.74	−0.16 (−0.9)	+0.43 (+1.2)
Fashion T=20	2	62.61	62.72	+0.10 (+0.4)	−0.18 (−0.3)
Fashion T=20	5	49.79	50.23	+0.44 (+1.5)	−0.33 (−0.7)

Table 4: Causal probe, $T = 10$, $\lambda = 1000$: paired differences vs. $b = 0$ ($n = 10$ paired seeds). †: Holm-corrected $p < 0.05$ within the $\Delta\ell$ family; ‡: within the Δavg family; §: within the ΔBWT family. Intervals are 95% CI for the paired mean difference; d_z is the paired standardized effect size.

dataset	arm (b)	$\Delta\ell$	Δavg	ΔBWT	$p_{\text{Holm}}(\Delta\ell)$	d_z	95% CI
fashion	branch (1)	+0.94	+2.08	+1.26	0.247	+0.67	$[-0.06, +1.94]$
fashion	branch (3)	+0.81	+1.79	+1.09	0.248	+0.54	$[-0.27, +1.88]$
fashion	branch (10)	+0.97	+1.03	+0.06	0.247	+0.64	$[-0.11, +2.06]$
fashion	neuron (1)	+0.87	+1.03	+0.18	0.248	+0.53	$[-0.31, +2.06]$
fashion	neuron (3)	+1.23	+2.56	+1.48	0.140	+0.86	$[+0.21, +2.25]$
fashion	neuron (10)	+1.39	+0.69	−0.77	0.192	+0.77	$[+0.09, +2.68]$
mnist	branch (1)	+2.18†	+3.29‡	+1.23§	0.000	+2.48	$[+1.55, +2.81]$
mnist	branch (3)	+2.90†	+4.30‡	+1.57§	0.000	+3.73	$[+2.34, +3.45]$
mnist	branch (10)	+3.16†	+4.17‡	+1.13§	0.000	+3.33	$[+2.48, +3.84]$
mnist	neuron (1)	+1.99†	+3.40‡	+1.56§	0.000	+2.56	$[+1.44, +2.55]$
mnist	neuron (3)	+2.68†	+4.08‡	+1.56§	0.000	+4.21	$[+2.22, +3.13]$
mnist	neuron (10)	+3.32†	+4.80‡	+1.65§	0.000	+4.64	$[+2.81, +3.83]$

Table 5: Causal probe, $T = 20$, $\lambda = 1000$: paired differences vs. $b = 0$ ($n = 10$ paired seeds). †: Holm-corrected $p < 0.05$ within the $\Delta\ell$ family; ‡: within the Δavg family; §: within the ΔBWT family. Intervals are 95% CI for the paired mean difference; d_z is the paired standardized effect size.

dataset	arm (b)	$\Delta\ell$	Δavg	ΔBWT	$p_{\text{Holm}}(\Delta\ell)$	d_z	95% CI
fashion	branch (3)	+1.44†	+1.98‡	+0.57	0.040	+1.14	[+0.54, +2.34]
fashion	branch (10)	+2.18†	+1.96‡	−0.24	0.004	+1.73	[+1.28, +3.09]
fashion	neuron (1)	+1.78†	+1.58‡	−0.21	0.025	+1.29	[+0.79, +2.77]
fashion	neuron (3)	+2.19†	+2.09‡	−0.11	0.003	+1.84	[+1.34, +3.04]
fashion	neuron (10)	+1.94†	+1.03‡	−0.95	0.004	+1.72	[+1.13, +2.75]
mnist	branch (3)	+3.66†	+3.03‡	−0.66	0.000	+2.95	[+2.77, +4.55]
mnist	branch (10)	+4.13†	+3.48‡	−0.68	0.000	+2.46	[+2.93, +5.33]
mnist	neuron (1)	+2.14†	+2.78‡	+0.68	0.031	+1.22	[+0.89, +3.39]
mnist	neuron (3)	+3.72†	+3.25‡	−0.49	0.000	+2.36	[+2.59, +4.84]
mnist	neuron (10)	+4.56†	+3.81‡	−0.79	0.000	+2.57	[+3.29, +5.83]

Table 6: ConvADNet dose–response, paired vs. $b = 0$ ($n = 10$; Holm-corrected p within each metric family; 95% CI for the paired mean difference; d_z the paired standardized effect size). No cell survives correction on any metric, which is the pre-registered G1 criterion evaluated conservatively.

cell	arm (b)	$\Delta\ell$	Δavg	ΔBWT	$p_{\text{Holm}}(\Delta\ell)$	d_z	95% CI
fashion $T = 10$	1	-0.06	$+0.22$	$+0.31$	1.000	-0.22	$[-0.25, +0.14]$
fashion $T = 20$	1	$+0.31$	$+0.66$	$+0.36$	0.081	$+1.11$	$[+0.11, +0.51]$
mnist $T = 10$	1	$+0.17$	-1.64	-2.02	1.000	$+0.12$	$[-0.87, +1.22]$
mnist $T = 20$	1	$+0.63$	-0.24	-0.92	1.000	$+0.24$	$[-1.21, +2.48]$
fashion $T = 10$	3	$+0.02$	-0.89	-1.01	1.000	$+0.03$	$[-0.48, +0.52]$
fashion $T = 20$	3	$+0.29$	-0.51	-0.84	0.540	$+0.72$	$[+0.00, +0.58]$
mnist $T = 10$	3	$+0.30$	-0.99	-1.43	1.000	$+0.18$	$[-0.87, +1.47]$
mnist $T = 20$	3	$+1.46$	-0.26	-1.81	0.642	$+0.67$	$[-0.11, +3.02]$
fashion $T = 10$	10	$+0.36$	-2.29	-2.95	1.000	$+0.45$	$[-0.21, +0.94]$
fashion $T = 20$	10	$+0.31$	-0.87	-1.24	0.642	$+0.66$	$[-0.03, +0.64]$
mnist $T = 10$	10	$+0.07$	-0.84	-1.01	1.000	$+0.04$	$[-1.23, +1.38]$
mnist $T = 20$	10	$+1.28$	$+0.30$	-1.04	0.682	$+0.61$	$[-0.22, +2.79]$

Table 7: The three configurations at T=10 (n=10; paired vs. each configuration's own b=0). "dead" = mean dead-gate fraction; "HR$_\ell$" = 100 − ℓ_0. The repair's $\Delta\ell$ tracks baseline weakness, not mortality.

dataset	config	ℓ_0	HR$_\ell$	dead	$\Delta\ell_{b=10}$	Δavg$_{b=10}$
MNIST	bare-MLP	74.3	25.7	0.572	+3.32*	+4.80*
MNIST	GN-MLP	93.5	6.5	**0.765**	+0.57*	−3.88*
MNIST	ConvADNet	84.1	15.9	**0.000**	+0.07	−0.84
Fashion	bare-MLP	56.8	43.2	0.603	+1.39*	+0.69
Fashion	GN-MLP	79.3	20.7	**0.855**	+0.36*	−6.55*
Fashion	ConvADNet	75.3	24.7	**0.000**	+0.36	−2.29*

Table 8: Width × normalization factor scan (n=10; paired vs. each cell's own b=0; b=10 learning-term gain with paired t). The GN collapse appears at both widths; the width effect on the plain trunk has the opposite sign.

dataset	config	ℓ_0	$\Delta\ell_{b=10}$
MNIST	n_h=256, plain	74.3	+3.32* (t=14.7)
MNIST	n_h=256, GN	88.0	+0.71* (t=7.5)
MNIST	n_h=2000, plain	77.1	+4.31* (t=12.5)
MNIST	n_h=2000, GN	93.4	+0.58* (t=10.1)
Fashion	n_h=256, plain	56.8	+1.39* (t=2.4)
Fashion	n_h=256, GN	72.4	−0.16 (t= − 0.8)
Fashion	n_h=2000, plain	60.0	+3.45* (t=9.6)
Fashion	n_h=2000, GN	79.2	+0.49* (t=4.5)

Table 9: Final average accuracy under worst-case context flips (n=10 seeds; noise applied to the final-phase evaluation only; λ=1000 enforced, n_h=256 in all arms).

dataset	arm	ρ=0	ρ=0.1	ρ=0.2	ρ=0.4
MNIST	base (b=0)	63.86	63.79	63.72	63.62
MNIST	repaired (b=10)	68.67	68.67	68.67	68.67
Fashion	base (b=0)	42.20	42.15	42.09	42.04
Fashion	repaired (b=10)	42.89	42.89	42.89	42.89

Table 10: Four-arm control experiment ($n = 10$ paired seeds, $\rho = 0$, matched width $n_h = 256$, matched usage-weighted anchors). Differences are paired against the base arm of the same dataset and seed; p is Holm-corrected within each metric family; d_z is the paired standardized effect size.

dataset	arm	$\Delta\ell$	p	d_z	Δavg	p
fashion	const	-0.33	1.000	-0.15	$+0.49$	0.969
fashion	repaired	$+1.39$	0.154	$+0.77$	$+0.69$	0.526
fashion	nogate	$+1.02$	0.461	$+0.49$	$+2.07$	0.146
mnist	const	$+0.07$	1.000	$+0.08$	-0.25	0.969
mnist	repaired	$+3.32*$	0.000	$+4.64$	$+4.80*$	0.000
mnist	nogate	$+2.61*$	0.000	$+2.46$	$+3.64*$	0.000